Apply filters to SQL queries

Project description

In this activity, I used SQL to filter and retrieve specific information from organisational databases. Working with employee records and login attempt data, I applied SQL filtering techniques to identify relevant information based on dates, times, departments, office locations and login outcomes. The aim was to narrow down large datasets and make it easier to identify records relevant to a particular investigation or business requirement.

Retrieve after hours failed login attempts

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = 0;
```

I used SQL filtering with the `WHERE` clause and the `AND` operator to identify unsuccessful login attempts made after 18:00. This allowed me to narrow down the login records and identify potentially suspicious activity for further investigation.

Retrieve login attempts on specific dates

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

I used the `SELECT` statement to retrieve all columns from the `log_in_attempts` table. I then applied the `WHERE` clause to filter the records by date.

I used the `OR` operator to retrieve login attempts from either 8 May 2022 or 9 May 2022. This meant that records matching either date would be included in the results.

Retrieve login attempts outside of Mexico

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'Mex%';
```

I used the `SELECT` statement to retrieve all columns from the `log_in_attempts` table. I then applied the `WHERE` clause with the `NOT` and `LIKE` operators to filter the results.

- `WHERE` – specifies the filtering condition.
- `NOT` – excludes records matching the condition.
- `LIKE` – searches for a particular text pattern.
- `'Mex%'` – searches for country names beginning with "Mex". The `%` wildcard represents any number of characters.

Retrieve employees in Marketing

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' and office LIKE 'East%';
+-------------+-------------+----------+------------+----------+
| employee_id | device_id   | username | department | office   |
+-------------+-------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+-------------+----------+------------+----------+
7 rows in set (0.020 sec)
```

I used the `SELECT` statement to retrieve all columns from the `employees` table. I then applied two filtering conditions using the `WHERE` clause.

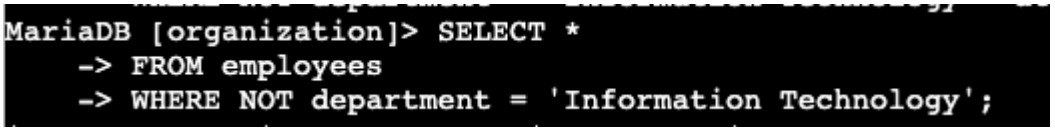
- `department = 'Marketing'` – selects employees working in the Marketing department.
- `AND` – ensures both conditions must be met.
- `office LIKE 'East%'` – identifies offices beginning with the word 'East'.
- `%` – acts as a wildcard, allowing any characters to follow 'East'.

Retrieve employees in Finance or Sales

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance'
    -> OR department = 'Sales';
```

I used the `WHERE` clause with the `OR` operator to retrieve employees working in either the Finance or Sales department.

Retrieve all employees not in IT



I used the `SELECT` statement to retrieve all columns from the `employees` table. I then applied the `WHERE` clause with the `NOT` operator to exclude employees working in the Information Technology department.

- `WHERE` – specifies which records to filter.
- `NOT` – reverses the condition.
- `department = 'Information Technology'` – identifies the department to exclude.

Summary

Throughout this activity, I practised using SQL filtering techniques, including `WHERE`, `AND`, `OR`, `NOT` and `LIKE`, as well as wildcard characters to search for specific patterns. I used these commands to investigate login attempts, filter employee records by department and location, and exclude information that was not relevant to the query. These skills are important in cybersecurity because they allow analysts to examine datasets, investigate potentially suspicious login activity and identify relevant information when reviewing organisational systems.